

TM Forum Component

Resource Catalog Management

TMFC010

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Notice

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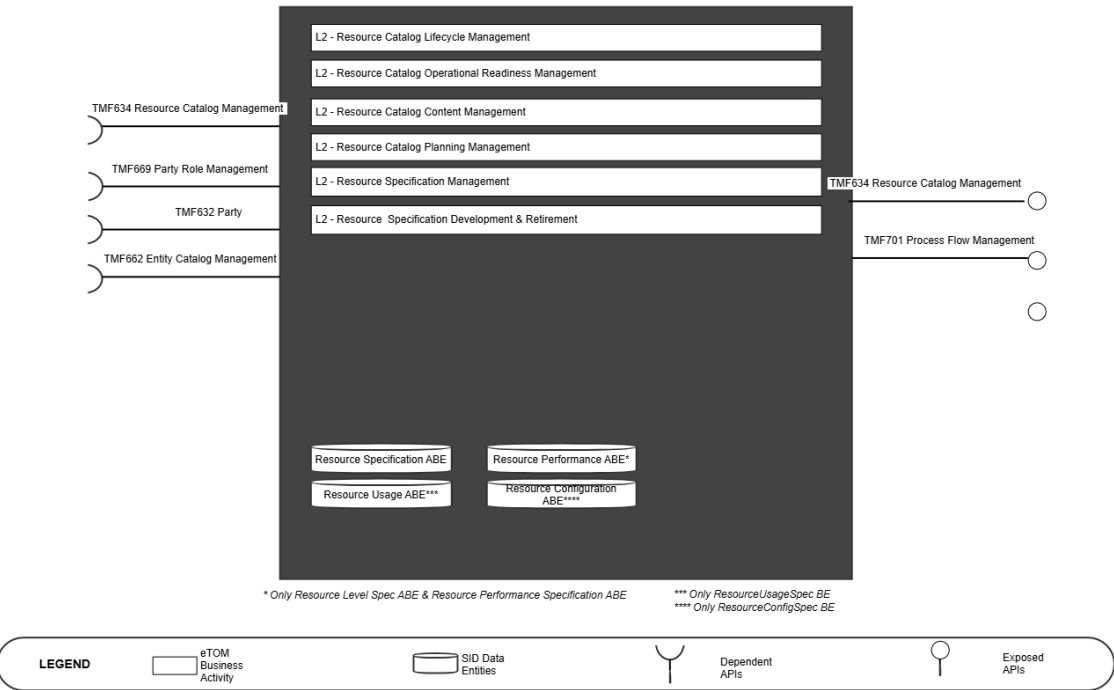
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1. Overview

Component Name	ID	Description	ODA Function Block
Resource Catalog Management	TMFC010	The Resource Catalog Management component is responsible for organizing the collection of resource specifications that identify and define all requirements for a resource. The Resource Catalog Management component has the functionality that enables presentation of a customer-facing view, so users are able to browse and select resources they need, as well as a technical view to enable definition and setup of resources contained in the resource catalog. Additional functionalities include capturing new resource specifications, managing resources (registering assets and components and identifying and mapping connections / relationships), administering the lifecycle of resources, describing relationships between resources, reporting on resources and changes to their attributes, and facilitating easy access to identify and assign resources.	Production

TMFC010 : Resource Catalog Management



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.5.15	L2	<i>Resource Catalog Lifecycle Management</i>	Catalog Lifecycle Management business process covers a set of business activities that enable manage the lifecycle of an organizations catalog from design to build according to defined requirements. Catalog Lifecycle Management proves the overarching governance to manage all the stages in the realization and operationalization of the Product/Service/Resource Catalog in support of the organizations business goals.
1.5.16	L2	<i>Resource Catalog Operational Readiness Management</i>	Resource Catalog Operational Readiness Management business process establishes and administers the support needed to operationalize Resource catalogs for ongoing day-to-day business needs. These business activities implement the Resource Catalog through Release and Deploy business activities.
1.5.17	L2	Resource Catalog Content Management	Resource Catalog Content Management business process define and provide the business activities that support the day-to-day operations of Resource Catalogs in order to realize the business operations goals. Resource Catalog Content Management business processes include administering the Resource Catalog instance in production, maintaining catalog entries, assuring catalogs, managing catalog access, managing entry lifecycle through versioning, handling catalog entity entry and changes, supporting distribution of catalogs as needed, and supporting user-facing activities.

Identifier	Level	Business Activity Name	Description
1.5.18	L2	Resource Catalog Planning Management	Resource Catalog Planning Management business process covers a set of business activities that understand and enable establish the plan to define, design and operationalize a catalog in order to meet the needs and objectives of Resource cataloging. The Resource Catalog Planning Management business process ensure that the organization is able to identify the most appropriate scheme and goal for it catalog. It includes designing the Catalog plan and developing the specification according to Resource management requirement.
1.5.19	L2	Resource Specification Management	Resource Specification Management business process leverages captured resource requirements to develop, master, analyze, and update documented standard conditions that must be satisfied by a resource design and/or delivery. Resource Specifications Management can result in establishing in a centralized way, technical (know-how) standards. Such standards provide the organization with a means to control and approve the values and inputs of specification through structure, review, approval and distribution processes to stakeholders and suppliers.
1.5.3	L2	Resource Specification Development & Retirement	Resource Specification Development & Retirement processes develop new, or enhance existing technologies and associated resource types, so that new Products are available to be sold to customers. They use the capability definition or requirements defined by Resource Strategy & Planning They also decide whether to acquire resources from outside, taking into account the overall business policy in that respect. These processes also retire or remove technology and associated resource types, which are no longer required by the enterprise. Resource types may be built, or in some cases leased from other parties. To ensure the most efficient and effective solution can be used, negotiations on network level agreements with other parties are paramount for both building and leasing. These processes interact strongly with Product and Engaged Party Development processes.
1.5.3.4	L3	Develop Detailed Resource Specifications	The Develop Detailed Resource Specifications processes develop and document the detailed resource-related technical, performance and operational specifications, and manuals. These

Identifier	Level	Business Activity Name	Description
			processes develop and document the required resource features, the specific technology requirements and selections, the specific operational, performance and quality requirements and support activities, any resource specific data required for the systems and network infrastructure. The Develop Detailed Resource Specifications processes provide input to these specifications. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally, the processes ensure that the documentation is captured in an appropriate enterprise repository.

2.2. SID ABEs

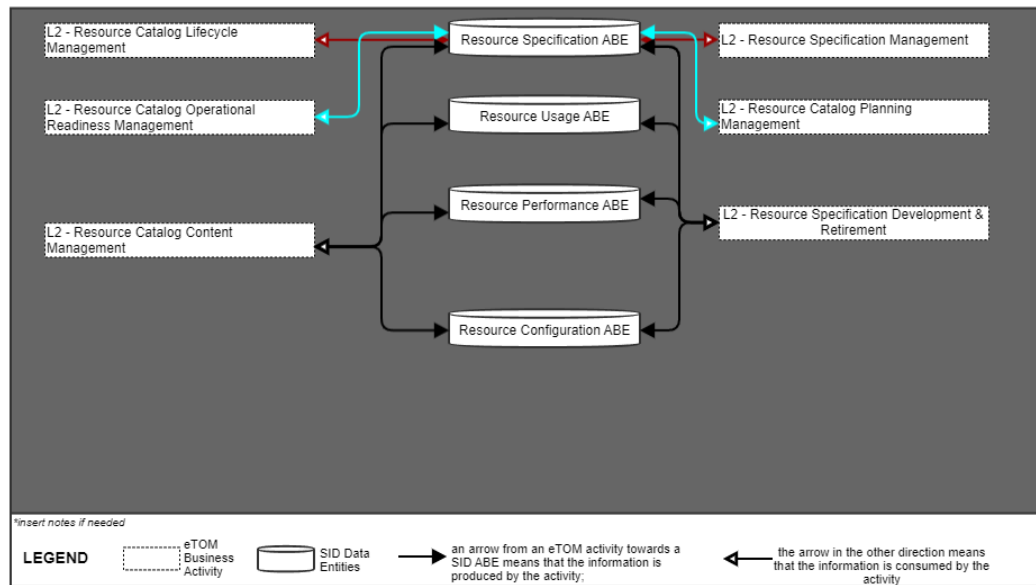
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Resource Specification ABE	
Resource Performance ABE	Resource Performance Specification BE
Resource Usage ABE	ResourceUsageSpec BE
Resource Configuration ABE	ResourceConfigSpec BE

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
737	Resource Capability Specification Management	This function involves the creation, editing, storage and retrieval of capability specifications. The capability specifications represent the general, common, and invariant characteristics of resource that may be realized in more than one type of specific resource. Examples of capability are Layer2, Data, radio, and Transport.	Resource Capability Management	Resource Specification Capability Development
467	Resource Data Transformation / Parsing Rules Configuration	Resource Data Transformation/Parsing Rules Configuration provides tools to set up and maintain resource data parsing rules	Resource Specification Management	Resource Specification Development
951	Resource Catalog Entities Management	Resource Catalog Entities Management identifies resource entities in a common Catalog Management from the Common Domain, or identifies a specific instance of a	Resource Specification Management	Resource Specification Development

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		Catalog Management for resource entities		
996	Resource Task Item Policy Control Configuration	Defines and configures the policies which will be implemented during the Resource task item lifecycle.	Resource Specification Management	Resource Specification Development
1083	Resource Specification Repository Management	Resource Specification Repository Management is able to create, modify and delete Resource Specification. This includes the ability to manage the state of an entity during its lifecycle (e.g. planned, deployed, in operation, replaced by locked). It includes Resource Specifications retrieval, integrity rules check and versioning management. It also provides Product Specification and Offering views adapted to the different roles.	Resource Specification Management	Resource Specification Development
1088	Resource Specification to Supplier Product Specification Relationship Design	Resource Specification to Supplier Product Specification Relationship Design identifies, when it corresponds to equipment rented to a Supplier (devices, network equipments, hardware & software, etc).	Resource Specification Management	Resource Specification Development
1089	Resource Specification Action Skill Design	Resource Specification Action Skill Design manages the links to the Skill catalog to identify for each type of resource, or of Action on a resource, which type of skill is necessary to make the intervention.	Resource Specification Management	Resource Specification Development
1082	Resource Specification Change Auditing	Resource Specification Change Auditing manages the implications of Resource Specifications changes to determine the consequences of any given change. Resource Specifications changes may	Resource Specification Management	Resource Specification Development

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		impact other Resource Specifications and Resource Facing Service Specit supports The function logs Resource Specifications changes and supports the analysis of relationships between Resource Specifications. In addition, it tracks the history of changes in an easy and accessible manner.		
1064	Logical and Software Resources Designing	Logical and Software Resources Designing supports physical, logical, and software design of resources including definition of configuration variables and initial parameters.	Resource Specification Management	Resource Specification Development
1087	Resource Specification Design	Resource Specification Design manages Resource Specifications including constraints, characteristics, and type of usages. It specifies Physical, Logical and Compound Resource Specifications. It includes resource types that as a Service Provider we don't own or commercialize (ex: mobile phones used by our customers).	Resource Specification Management	Resource Specification Development

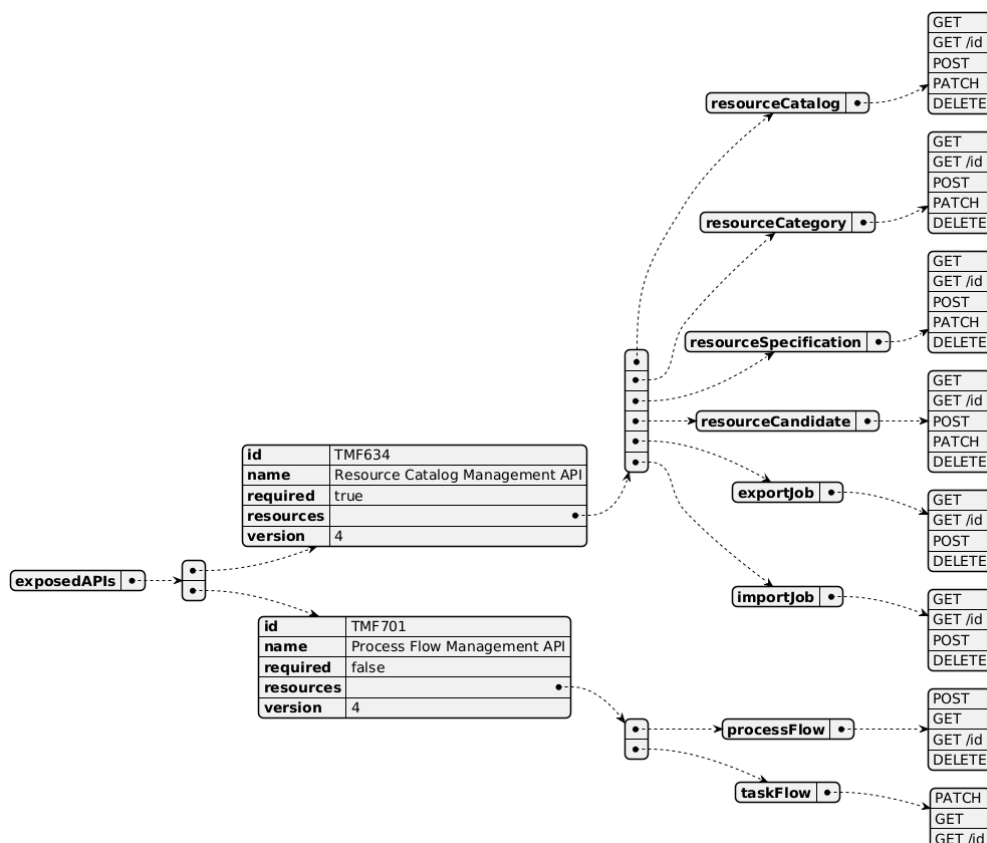
3. TM Forum Open APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

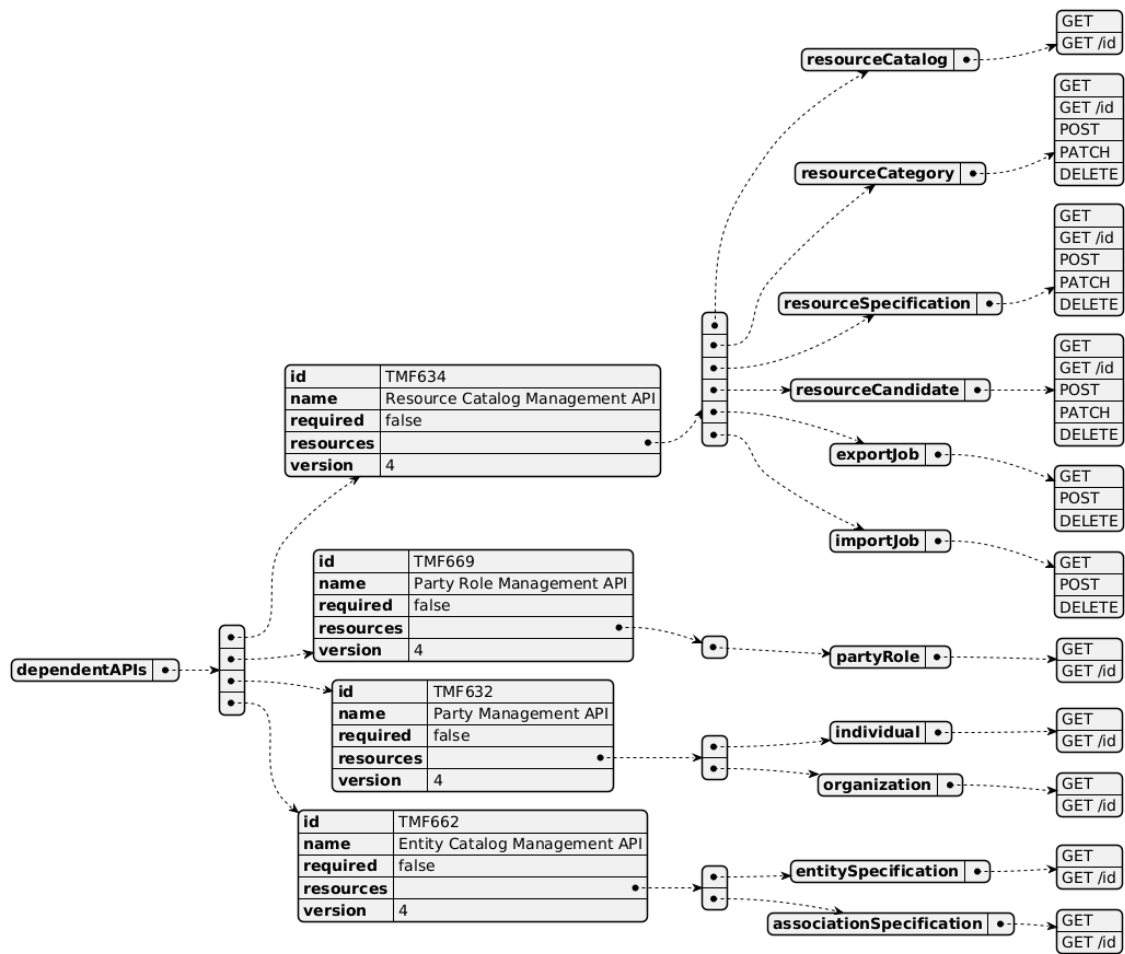
Following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	Operations
TMF634	Resource Catalog Management	4	Mandatory	<i>resourceCatalog</i> : POST, PATCH, GET, GET /id, DELETE <i>resourceCategory</i> : POST, PATCH, GET, GET /id, DELETE <i>resourceSpecification</i> : POST, PATCH, GET, GET /id, DELETE <i>resourceCandidate</i> : POST, PATCH, GET, GET /id, DELETE <i>exportJob</i> : POST, GET, GET /id, DELETE <i>importJob</i> : POST, GET, GET /id, DELETE
TMF701	Process Flow Management	4	Optional	<i>processFlow</i> : POST, GET, GET /id, DELETE <i>taskflow</i> : PATCH, GET, GET /id

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the resource catalog component:



API ID	API Name	API Version	Mandatory / Optional	Operations
TMF634	Resource Catalog Management	4	Optional	<i>resourceCatalog: GET, GET /id resourceCategory: POST, PATCH, GET, GET /id, DELETE resourceSpecification: POST, PATCH, GET, GET /id, DELETE resourceCandidate: POST, PATCH, GET, GET /id, DELETE exportJob: POST, GET, DELETE importJob: POST, GET, DELETE</i>
TMF669	Party Role Management	4	<i>Optional</i>	<i>partyRole: GET, GET /id</i>
TMF632	Party	4	<i>Optional</i>	<i>individual: GET, GET /id organization: GET, GET /id</i>
TMF662	Entity Catalog Management	4	<i>Optional</i>	<i>entitySpecification: GET, GET /id associationSpecififaction: GET, GET /id</i>

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

5.2. Jira References

5.2.1. Open API

 [AP-4088](#) - Assess the addition of a ResourceUsage API with both ResourceUsage & ResourceUsageSpecification **DONE**

The UsageSpecification is described in the Catalog Management components – at product, service, and resource levels.

As part of the Resource Catalog Component specification, the question has arisen around the need to assess the addition of a ResourceUsage API with both ResourceUsage & ResourceUsageSpecification (feedback from Ludovic Robert and Kamal Maghsoudlou). This will be similar to the resource level like TMF727 (ServiceUsage) is for the service level. This is consistent with the SID ABE (resourceUsage & resourceUsageSpec).

5.2.2. Functional Framework

 [ISA-996](#) - Master Data Management has repurposed catalog related functions **BACKLOG**

As part of reviewing the Product, Service and Resource Catalogs to align with the updates to the Functional Framework, we realized that a lot of functions have either been removed or been re-purposed by Master Data Management.

Due to this, we had to make a decision to remove all these functions from the Components:

- Removed deleted unclassified functions (1,2,5,7,9,10, 12,14,15)
- Removed functions that used to be related to Catalog management but now have been re-purposed by Master Data Management (3,4, 6,11,13)

Our concern is that we may be missing functions for the catalog due to re-use of Master Data Management.

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	05-Aug-2022	Kamal Maghsoudlou Gaetano Biancardi Sylvie Demarest	Final edits prior to publication
1.1.0	06-Oct-2022	Elisabeth Andersson	Added support for federated catalogs and minor fixes.
1.1.1	27 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.1.1	14-Aug-2023	Amaia White	Final edits prior to publication
1.2.0	19 Apr 2024	Elisabeth Andersson	• Update to latest template • Aligned with 23.5 SID and eTOM - remove of Resource Test specification BE and process (to be managed by Resource Test Mgt) • Updated with Functional Frameworks 23.5 ○ Added new functions under lvl2 Resource Specification Development (1064, 1082, 1083, 1087, 1088,1089) ○ Updated the description of function 996 under lvl2 Resource Specification Development ○ Removed deleted unclassified functions (1,2,5,7,9,10, 12,14,15) ○ Removed functions that used to be related to

Version Number	Date Modified	Modified by:	Description of changes
			Catalog management but now have been re-purposed by Master Data Management (3,4,6,11,13)
1.2.0	30 Apr 2024	Amaia White	Final edits prior to publication
1.3.0	26 Nov 2024	Elisabeth Andersson	- TMF688, removed from the core specification, moved to supporting functions - TMF672, removed from the core specification, moved to supporting functions - Removed the minor versions of all APIs as per template
1.3.0	24 Dec 2024	Rosie Wilson	Final edits prior to publication
1.3.1	08 Jul 2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025
1.3.1	15 Jul 2025	Rosie Wilson	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	05-Aug-2022	Goutham Babu	Initial release
Pre-production	07-Oct-2022	Alan Pope	Version 1.1.0
Pre-production	14-Aug-2023	Amaia White	Version 1.1.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	30-Apr-2024	Amaia White	Version 1.2.0
Production	28-Jun-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	24-Dec-2024	Rosie Wilson	Version 1.3.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	15 Jul 2025	Rosie Wilson	Version 1.3.1

6.2. Acknowledgements

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